Modelling impacts of climate change on cattle production

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1. Analyse spatio-temporal changes in THI for cattle (1981–2098)

Research Question	Indicators	Data Source / Variable	Unit / Scale	Analysis Method	Temporal Resolution	Spatial Resolution	Expected Outcome	Challenges
a)What is the spatio-temporal pattern of variation in THI?	-THI values Heat stress days	-ERA5 -CORDEX-CORE Daily tasmax (°C) & RH (%)	% (THI)	Python:Xarray/Matplotlib/geopandas/rasterio -Mann-Kendall trend test -Sen's slope	Daily, monthly, seasonal, annual	25–31 km	Maps & graphs showing THI trends	Batch file processing: -memory issues working with python and laptop -clipping to the study area as a batch-coding -working with 2 regions: East Africa and Southern Africa
b) What are the shifts in number of heat stress days?	-Days exceeding THI thresholds	- THI	Days/year	-Time-series analysis - Hotspot mapping with geopandas & rasterio	Daily, seasonal, annual	25–31 km	Identify periods/regions with increasing heat stress days	-Results come as number of grids not unique days

2. Assess spatio-temporal dynamics of NPP and vegetation facet cover from 1981 to 2098

Research Question	Indicators	Data Source / Variable	Unit / Scale	Analysis Method	Temporal Resolution	Spatial Resolution	Expected Outcome	Challenges
a) How have NPP & vegetation-facet cover changed 1981–2098?	-NPP (gC/m²/year) - % grass, shrub, tree cover	-AF-RANGE outputs TIPPECC climate data integration in AF_Range: Tmin, Tasmax, Precip/month/RCP 2.6 & 8.5	gC/m²/year (NPP), % cover	-Mann-Kendall trend test Sen's slope -GIS mapping via geopandas	Monthly, annual	20 km	Maps showing NPP & vegetation trends (historical & future)	Integration of TIPPEC climate data: -Resampling from 25km to 20km -Reprojecting Validation of AF_Range? MODIS NPP? -Re-griding: 500m to match Af_Range grid 10.167°
b) How do compound climate extremes affect forage stability?	-Biomass persistence -"false good years"	-AF-RANGE -ERA5 & CORDEX-CORE :precipitation & temp	Days/year, % change in NPP	-Pandas/xarray for sequence analysis -Matplotlib for temporal patterns	Seasonal	20–25 km	Identify years/regions vulnerable to forage collapse	
c) What relationships exist between NPP, vegetation-facet cover & SPEI?		-NPP -grass/tree/shrub cover -SPEI	-AF-RANGE -SPEI datasets	gC/m²/year, % cover, SPEI index	-Multiple linear regression -Spatial correlation	Seasonal, annual	20–25 km	

3. Determine future suitability index of cattle production

Research Question	Indicators	Data Source / Variable	Unit / Scale	Analysis Method	Temporal Resolution	Spatial Resolution	Expected Outcome	Challenges
a)Which shifts will occur in the suitability niches for cattle production areas ?	- Suitability score (1–5)	-AF-RANGE NPP -THI -Precipitation -Slope -Soil type -LULC	Suitability class (1–5) %	- Multi-Criteria Decision Analysis + AHP - Scenario mapping	Seasonal, annual	Different	Identify areas losing or gaining suitability.	Different resolutions
b) When will a tipping point be reached under RCP 2.6 and RCP 8.5 in land suitability?	Suitability Index crossing threshold			- Time series analysis, threshold detection, scenario comparison	Annual		Map of tipping point years, identification of early-risk areas, comparison between RCPs	
b) Which areas will become hotspots of vulnerability?	- Combined index of THI, NPP, SPEI, cattle density	- AF-RANGE - CORDEX-CORE - GLW	High/Medium/Low risk	- GIS hotspot analysis (rasterio, geopandas)	Seasonal, annual	20–25 km	Maps showing vulnerable areas; guide interventions	